Math 240 Matlab Project 3

Table of Contents

Problem 1 1
Problem 2 3
Problem 3 4
Problem 4 5
Problem 5 6
Problem 6 8

Fall 2024

Name: Aditya Agrawal

*Type/paste Matlab commands for a particular problem/part below the correspondingly labeled line, above the next double % sign. Written responses to questions (not lines of code) should be given as comments, following a single percent symbol % format rat

Problem 1

```
v1 = [-6; 4; -9; 4]
v2 = [8; -3; 7; -3]
v3 = [-9; 5; -8; 3]
x = [4; 7; -8; 3]

V = [v1 v2 v3]
rref(V)
% Since v1, v2, and v3 are linearly independent, B is a basis for H

A = [V, x]
rref(A)
% Since v1, v2, v3, and x are not linearly independent, x is in H

x = [3; 5; 2; 0]
```

```
v1 =

      -6
       4
      -9
       4


v2 =

       8
      -3
       7
      -3
```

```
v3 =

    -9
     5
    -8
     3


x =

     4
     7
    -8
     3


V =

    -6     8    -9
     4    -3     5
    -9     7    -8
     4    -3     3


ans =

     1     0     0
     0     1     0
     0     0     1
     0     0     0


A =

    -6     8    -9     4
     4    -3     5     7
    -9     7    -8    -8
     4    -3     3     3


ans =

     1     0     0     3
     0     1     0     5
     0     0     1     2
     0     0     0     0


x =

     3
     5
```

```
     2
     0
```

Problem 2

```
format short
% (a)
A = [1 cos(0) cos(0)^2 cos(0)^3;
    1 cos(0.1) cos(0.1)^2 cos(0.1)^3;
    1 cos(0.2) cos(0.2)^2 cos(0.2)^3;
    1 cos(0.3) cos(0.3)^2 cos(0.3)^3]


A =

    1.0000    1.0000    1.0000    1.0000
    1.0000    0.9950    0.9900    0.9851
    1.0000    0.9801    0.9605    0.9414
    1.0000    0.9553    0.9127    0.8719
```

(b)

```
rref(A)
det(A)


ans =

     1     0     0     0
     0     1     0     0
     0     0     1     0
     0     0     0     1


ans =

   6.5176e-11
```

(c) Because there is a pivot in every row of the rref of A, it is invertible Because the determinant of A is not 0, A is invertible

(d)

```
A = [1 cos(0) cos(0)^2 cos(0)^3;
    1 cos(0.2) cos(0.2)^2 cos(0.2)^3;
    1 cos(0.5) cos(0.5)^2 cos(0.5)^3;
    1 cos(1) cos(1)^2 cos(1)^3]

rref(A)
det(A)


A =
```

```
    1.0000    1.0000    1.0000    1.0000
    1.0000    0.9801    0.9605    0.9414
    1.0000    0.8776    0.7702    0.6759
    1.0000    0.5403    0.2919    0.1577


ans =

     1     0     0     0
     0     1     0     0
     0     0     1     0
     0     0     0     1


ans =

   1.7052e-05
```

(e) The functions {1, sin^2(t), cos^2(t)} are linearly dependent because they satisfy the Pythagorean identity: sin^2(t) + cos^2(t) = 1. This implies one function can be written as a combination of the others.

Problem 3

```
format rat
% (a)
A = [1 0 0 0 0 0 0;
    0 1 0 0 0 0 0;
    -1 0 2 0 0 0 0;
    0 -3 0 4 0 0 0;
    1 0 -8 0 8 0 0;
    0 5 0 -20 0 16 0;
    -1 0 18 0 -48 0 32]
rref(A)
% Set C is linearly independent because each colomn has a pivot


A =

  Columns 1 through 5

       1             0             0             0             0
       0             1             0             0             0
      -1             0             2             0             0
       0            -3             0             4             0
       1             0            -8             0             8
       0             5             0           -20             0
      -1             0            18             0           -48

  Columns 6 through 7

       0             0
       0             0
```

```
     0     0
     0     0
     0     0
    16     0
     0    32


ans =

  Columns 1 through 5

     1     0     0     0     0
     0     1     0     0     0
     0     0     1     0     0
     0     0     0     1     0
     0     0     0     0     1
     0     0     0     0     0
     0     0     0     0     0

  Columns 6 through 7

     0     0
     0     0
     0     0
     0     0
     0     0
     1     0
     0     1
```

(b) Since we have shown that C is linearly independent, and we have 7 vectors in C, which matches the dimension of H, it follows that C spans H. Therefore, C is a basis for H.

Problem 4

(a)

```
format rat
A = [2 3 1 8 1; -19 -12 22 11 -2; -7 -5 7 1 -1; -10 -4 16 18 0];
A
rank(A)


A =

     2     3     1     8     1
   -19   -12    22    11    -2
    -7    -5     7     1    -1
   -10    -4    16    18     0


ans =
```

```
       2
```

(b) dim(Col A) = 2 dim(Row A) = 2 dim(Nul A) = 3

(c)

```
rref(A)
% Basis of Col A = {[2;-19;-7;-10] , [3;-12;-5;-4]}
% Basis of Row A = {[1 0 -26/11 -43/11 -2/11] , [0 1 21/11 58/11 5/11]}
% Basis of Nul A = {[26/11;-21/11;1;0;0] , [43/11;-58/11;0;1;0] ,
% [2/11;-5/11;0;0;1]}


ans =

       1            0         -26/11       -43/11        -2/11
       0            1          21/11        58/11         5/11
       0            0             0            0            0
       0            0             0            0            0
```

Problem 5

(a)

```
v1 = [1; 2; 1; 3]
v2 = [3; 3; 4; 7]
v3 = [3; 0; 5; 5]
v4 = [5; 1; -3; 8]
v5 = [1; -1; 2; 1]


v1 =

       1
       2
       1
       3


v2 =

       3
       3
       4
       7


v3 =

       3
       0
       5
```

```
       5


v4 =

       5
       1
      -3
       8


v5 =

       1
      -1
       2
       1
```

(b)

```
A = [v1 v2 v3 v4 v5]
rref_A = rref(A)
rank_A = rank(A)
[v1 v2 v4]


A =

       1       3       3       5       1
       2       3       0       1      -1
       1       4       5      -3       2
       3       7       5       8       1


rref_A =

       1       0      -3       0      -2
       0       1       2       0       1
       0       0       0       1       0
       0       0       0       0       0


rank_A =

       3


ans =

       1       3       5
       2       3       1
       1       4      -3
       3       7       8
```

(c) {p1(t);p2(t);p4(t)} = {1+2t+t^2+3t^3; 3+3t+4t^2+7t^3; 5+t−3t^2+8t^3} dim W = 3

(d) W is not equal to P3 because dim W = 3 and P3 = 4. W is a proper subspace of P3

Problem 6

(a)

```
A1 = [0 -1 2; 4 5 -2];
A2 = [-4 0 1; -3 0 2];
A3 = [-3 2 2; 4 -2 1];
A4 = [-12 -2 1; 2 3 2];

v1 = [0;-1;2;4;5;-2]
v2 = [-4;0;1;-3;0;2]
v3 = [-3;2;2;4;-2;1]
v4 = [-12;-2;1;2;3;2]


v1 =

       0
      -1
       2
       4
       5
      -2


v2 =

      -4
       0
       1
      -3
       0
       2


v3 =

      -3
       2
       2
       4
      -2
       1


v4 =

     -12
      -2
       1
```

```
       2
       3
       2
```

(b)

```
V = [v1 v2 v3 v4];
V
rank(V)
rref(V)
```

```
V =

       0            -4            -3           -12
      -1             0             2            -2
       2             1             2             1
       4            -3             4             2
       5             0            -2             3
      -2             2             1             2


ans =

       4


ans =

       1             0             0             0
       0             1             0             0
       0             0             1             0
       0             0             0             1
       0             0             0             0
       0             0             0             0
```

(c) Vectors are not linearly independent

Published with MATLAB® R2024b